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Inventor(s)

NISHIDA; Yuta et al.

TAPERED ROLLER BEARING

Abstract

Provided is a tapered roller bearing in which an inner ring assembly is integrated by a connector such as a wire and a connector can be removed after incorporating a bearing, and the tapered roller bearing can realize a roller guide. The tapered roller bearing removably includes a falling-off preventer to prevent a tapered roller contained in a pocket having a first guide claw extending toward an inner diameter side of a retainer segment from falling off toward the outer diameter side. The falling-off preventer is engaged with an outer diameter surface and a large-diameter-side lateral-surface of a large-diameter-side arc-shaped portion of the retainer segment. The retainer segment and the falling-off preventer are connected with the connector.

Inventors: NISHIDA; Yuta (Kuwana, JP), ITO; Tomohito (Kuwana, JP)

Applicant: NTN CORPORATION (OSAKA, JP)

Family ID: 91084826

Assignee: NTN CORPORATION (OSAKA, JP)

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Background/Summary

CROSS REFERENCE TO THE RELATED APPLICATION

[0001] This application is a continuation application, under 35 U.S.C. § 111 (a) of international patent application No. PCT/JP2023/040860, filed Nov. 14, 2023, which claims priority to a Japanese patent application No. 2022-185128 filed Nov. 18, 2022 and a Japanese patent application No. 2023-180301 filed Oct. 19, 2023, the entire disclosures of all of which are herein incorporated by reference as a part of this application.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to a tapered roller bearing for use as a main shaft of a wind power generator or in an industrial machinery, in particular, to a large tapered roller bearing, for example, whose outer diameter exceeds 1 m.

Description of Related Art

[0003] A tapered roller bearing generally includes a retainer made of steel sheet. The retainer made of steel sheet is produced, for example, through a press process. However, for a large tapered roller bearing whose outer diameter exceeds 1 m, it is difficult to be formed through the press process due to the problems on facilities. On the other hand, when the retainer is produced by cutting, the costs will increase and there will be more waste in view of material loss as compared with pressed products. To reduce the costs, a segment retainer has been produced by forming a plurality of retainer segments by injection molding of resin and integrating them (e.g., Patent Document 1). Incidentally, it is difficult to form a large integrated ring exceeding 1 m by injection molding of resin.

[0004] In such segment retainer, for example, when a user incorporates a single row tapered roller bearing into a wind turbine, there is a process for treating an inner ring assembly with a small end surface facing downward. In the process, to prevent rollers from falling off from the retainers, a method has been devised for integrating the segment retainers by a connector such as a wire to put together the retainers (for example, Patent Documents 2 to 4). [0005] [Patent Document 1] JP Patent No. 4342512 [0006] [Patent Document 2] JP Laid-open Patent Publication No. 2011-137530 [0007] [Patent Document 3] WO 2013/092217 [0008] [Patent Document 4] JP Laid-open Patent Publication No. 2018-080747

[0009] The problem of the segment retainer connected by the wire is that the wind turbine is driven with the wire attached to the bearing. Even if the wire is detached after assembling the wind turbine, the retainer is guided by a rolling contact surface of a roller, an inner ring raceway and an outer ring raceway, which is not a roller guide. If a conventional “roller guide segment retainer” in which the wire is not used is temporarily integrated by the wire, there is a possibility that the rollers fall off when the inner ring assembly is turned upside down. In the case where the wind turbine is driven with the wire attached, there is a fear that wear, abnormal sound, and other problems are generated due to the wire. In addition, in the case where the retainer is guided by the raceway, there is a possibility of wear of a contact part of the retainer with the raceway and the raceway.

SUMMARY OF THE INVENTION

[0010] An object of the present invention is to provide a tapered roller bearing in which an inner ring assembly is integrated by a connector such as a wire and the connector can be removed after incorporating a bearing, and the tapered roller bearing can realize a roller guide.

[0011] A tapered roller bearing according to the present invention includes: an inner ring; an outer ring; a plurality of tapered rollers interposed between the inner ring and the outer ring; and a retainer which retains the tapered roller, the retainer including a plurality of retainer segments divided in a circumferential direction, the plurality of the retainer segments being connected by a removable connector. Each of the retainer segments has three or more pockets arranged in the circumferential direction which hold the tapered rollers. Each of the retainer segments has a large-diameter-side arc-shaped portion and a small-diameter-side arc-shaped portion extending circumferentially and a plurality of pillars which connect the large-diameter-side arc-shaped portion and the small-diameter-side arc-shaped portion. The pockets consisting of: the large-diameter-side arc-shaped portion; the small-diameter-side arc-shaped portion; and the pillar. A part of the pockets has a first guide claw extending toward an inner diameter side from the pillar and the rest of the pockets has a second guide claw extending toward an outer diameter side from the pillar. In addition, the tapered roller bearing includes a removable falling-off preventer which prevents the tapered roller contained in the pocket having the first guide claw from falling off toward the outer diameter side. The falling-off preventer is engaged with an outer diameter surface and a large-diameter-side lateral-surface of the large-diameter-side arc-shaped portion of the retainer segment. The retainer segments and the falling-off preventer are connected with the connector. The connector is, for example, a wire. The retainer is, for example, a retainer for a large tapered roller bearing, for example, whose outer diameter exceeds 1 m.

[0012] According to this configuration, since the connector and the falling-off preventer are removable, the inner ring assembly is integrated by the connector and the falling-off preventer, and the connector can be removed after incorporating a bearing. Moreover, since the retainer is guided by only the tapered roller with the first guide claw and the second guide claw during operation, the roller guide can be realized.

[0013] In the present invention, the falling-off preventer may be inserted from a large end surface side of the inner ring and may be arranged along the pillar or the second guide claw. According to this configuration, the falling-off preventer can lead to suppress the movement of the tapered roller in a radial direction.

[0014] Further, in the present invention, each of the retainer segments may include, on an end portion in the circumferential direction of the large-diameter-side arc-shaped portion, a falling-off preventer guide portion which suppresses the movement of the falling-off preventer in the circumferential direction. According to this configuration, the falling-off preventer can be easily attached to and detached from the retainer segment.

[0015] In the present invention, the falling-off preventer may have a guide portion with which the connector is engaged. In this case, the guide portion may be an insertion hole into which the connector is inserted or an engagement groove with which the connector is engaged. According to this configuration, it is easy to integrate the retainer segments with the falling-off preventer by the connector.

[0016] In the present invention, the falling-off preventer may have an inner ring engaging portion engaged with a large end surface of the inner ring. According to this configuration, the inner ring engaging portion can prevent an inner-diameter side of the retainer segment from spreading due to tightening of the connector.

[0017] In the present invention, moreover, the tapered roller bearing may further include a large-diameter-side protruding portion which has an engaging portion with the connector and is disposed on opposite end sides of the large-diameter-side lateral-surface of the large-diameter-side arc-shaped portion of the retainer segment. According to this configuration, it is easy to integrate the retainer segments with the falling-off preventer by the connector.

[0018] In the present invention, opposite ends of the connector may be connected together by a fastening portion or a fastening member. According to this configuration, it is easy to handle the connector.

[0019] In the present invention, the connector may be divided into a plurality of connector segments, and ends of each of the connector segments may be connected together by a fastening portion or a fastening member. In this case, a plurality of the fastening portions or the fastening members may be arranged at equal intervals in the circumferential direction. According this configuration, it is possible to apply uniform fastening force to the connector by the fastening portion or the fastening member.

[0020] In the present invention, moreover, the tapered roller bearing may further include a small-diameter-side protruding portion, into which a ring-shaped jig can be fitted, which portion is arranged on a small-diameter-side lateral-surface of the small-diameter-side arc-shaped portion of the retainer segment. According to this configuration, the ring-shaped jig can prevent the small diameter side of the retainer segment arranged in a ring shape from spreading.

[0021] In the present invention, the retainer segment may be made of Poly Ether Ketone that contains carbon fibers or glass fibers. In addition, the falling-off preventer may be made of resin or metal.

[0022] In the present invention, in a state in which the tapered roller is arranged on a raceway surface of the inner ring with the large end surface of the inner ring facing downward, when an angle of a large flange portion of the inner ring with respect to a straight line that is perpendicular to a central axis of the inner ring is defined as I , a chamfered width of a leading edge of the large flange portion is defined as H , a distance from the central axis of the inner ring to a center of gravity of the tapered roller is defined as y_1 , and a diameter of the large flange portion is defined as J , the condition below may be satisfied.

$$[00001](J/2) - H \cdot \text{Math. cos} I > y_1$$

[0023] According to this configuration, when the tapered roller is incorporated into the inner ring, it is possible to prevent the tapered rollers from falling off due to self-weight.

[0024] In the present invention, in a state in which the retainer segments are connected by the connector, when a diameter of the small flange portion of the inner ring is defined as M , an angle of the small flange portion with respect to a straight line that is perpendicular to a central axis of the inner ring is defined as L , a chamfered width of a leading edge of the small flange portion is defined as K , and a distance, to the central axis, from a contact point C at which a side surface of the small flange portion comes into contact with an end surface of the tapered roller on the small diameter side when the tapered roller is rotated around a leading edge of the large flange portion of the inner ring is defined as y_3 , the condition below may be satisfied.

$$[00002](M/2) - K \cdot \text{Math. cos} L > y_3$$

[0025] According to this configuration, since the end surface of the tapered roller on the small diameter side comes into contact with the contact point C on the side surface of the small flange portion, the rotation of the tapered roller is suppressed.

[0026] Further, in the present invention, the tapered roller bearing may further include a small-diameter-side-engaging portion which removably engages a small-diameter-side-connector and is provided on the small-diameter-side lateral-surface of the small-diameter-side arc-shaped portion of the retainer segment. According to this configuration, the small-diameter-side-connector can prevent the inner diameter side of the retainer segments arranged in a ring shape from spreading.

[0027] Any combination of at least two constructions, disclosed in the appended claims and/or the specification and/or the accompanying drawings should be construed as included within the scope of the present invention. In particular, any combination of two or more of the appended claims should be equally construed as included within the scope of the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] In any event, the present invention will become more clearly understood from the following description of preferred embodiments thereof, when taken in conjunction with the accompanying drawings. However, the embodiments and the drawings are given only for the purpose of illustration and explanation, and are not to be taken as limiting the scope of the present invention in any way whatsoever, which scope is to be determined by the appended claims. In the accompanying drawings, like reference numerals are used to denote like parts throughout the several views:

[0029] FIG. 1 is a cross-sectional view illustrating a tapered roller bearing according to a first embodiment of the present invention;

[0030] FIG. 2 is a front view of an inner ring assembly of the tapered roller bearing as viewed from an axial direction;

[0031] FIG. 3 is a perspective view illustrating the inner ring assembly;

[0032] FIG. 4 is a cross-sectional view illustrating a retainer segment of a retainer of the tapered roller bearing;

[0033] FIG. 5 is a top plan view of the retainer segment as viewed from an outer side in a radial direction;

[0034] FIG. 6 is a perspective view illustrating the retainer segment;

[0035] FIG. 7 is a perspective view illustrating a variation of a falling-off preventer of the inner ring assembly;

[0036] FIG. 8 is a perspective view illustrating another variation of a falling-off preventer of the inner ring assembly;

[0037] FIG. 9 is a perspective view illustrating yet another variation of a falling-off preventer of the inner ring assembly;

[0038] FIG. 10 is a cross-sectional view illustrating an inner ring assembly in which the falling-off preventer is used;

[0039] FIG. 11 is a cross-sectional view illustrating an inner ring assembly in which the falling-off preventer is used;

[0040] FIG. 12 is a cross-sectional view illustrating the tapered roller bearing;

[0041] FIG. 13 is a schematic view illustrating a relationship between a tapered roller and a large flange portion of an inner ring when a tapered roller is arranged in the inner ring with the large end surface of the inner ring facing downward;

[0042] FIG. 14 is a perspective view illustrating tapered rollers inserted into the retainer segment;

[0043] FIG. 15 is a perspective view illustrating the retainer segment inserted into the inner ring;

[0044] FIG. 16 is a cross-sectional view illustrating the retainer segment inserted into the inner ring;

[0045] FIG. 17 is a perspective view illustrating the falling-off preventers inserted into the retainer segment;

[0046] FIG. 18 is a perspective view illustrating the retainer segments arranged in a ring shape;

[0047] FIG. 19 is a perspective view illustrating the inner ring assembly to which a ring-shaped jig and connectors are attached;

[0048] FIG. 20 is a schematic view illustrating a rotating state of the tapered roller arranged on a raceway surface of the inner ring;

[0049] FIG. 21 is a schematic view illustrating a relationship between a height of a small flange portion of the inner ring and the tapered roller when the tapered roller arranged on the raceway surface of the inner ring is rotated;

[0050] FIG. 22 is a cross-sectional view illustrating the inner ring assembly incorporated into an

outer ring;

[0051] FIG. **23** is a cross-sectional view illustrating the inner ring assembly incorporated into the outer ring;

[0052] FIG. **24** is a cross-sectional view illustrating a tapered roller bearing according to a second embodiment of the present invention;

[0053] FIG. **25** is a top plan view illustrating a segment of a segment retainer used in the tapered roller bearing of FIG. **24**;

[0054] FIG. **26** is a perspective view illustrating the segment of the segment retainer used in the tapered roller bearing of FIG. **24**;

[0055] FIG. **27** is a cross-sectional view illustrating the segment of the segment retainer used in the tapered roller bearing of FIG. **24**;

[0056] FIG. **28** is a cross-sectional view illustrating another example of a segment of a segment retainer used in the tapered roller bearing of FIG. **24**;

[0057] FIG. **29** is a perspective view illustrating the tapered roller bearing of FIG. **24** in which an outer ring is removed;

[0058] FIG. **30** is a front view illustrating the tapered roller bearing of FIG. **24** in which the outer ring is removed;

[0059] FIG. **31** is a cross-sectional view illustrating the tapered roller bearing of FIG. **24** after a connector is bound;

[0060] FIG. **32** is a cross-sectional view illustrating a third embodiment of a tapered roller bearing according to the present invention;

[0061] FIG. **33** is a perspective view illustrating a fourth embodiment of a tapered roller bearing according to the present invention;

[0062] FIG. **34** is a front view of an inner ring assembly of a tapered roller bearing according to a fifth embodiment of the present invention as viewed from an axial direction;

[0063] FIG. **35** is a perspective view of the inner ring assembly;

[0064] FIG. **36** is a cross-sectional view illustrating a retainer segment of a retainer of the tapered roller bearing;

[0065] FIG. **37** is a top plan view of the retainer segment as viewed from an outer side in a radial direction; and

[0066] FIG. **38** is a perspective view of the retainer segment.

DESCRIPTION OF EMBODIMENTS

[0067] Favorable embodiments of the present invention will be described in lights of Figures. In the following description, a direction along a bearing center axis AX is referred to as an “axial direction”, a direction perpendicular to the bearing center axis AX is referred to as a “radial direction”, and a circumferential direction around the bearing center axis AX is referred as a “circumferential direction”. In addition, a side toward the bearing center axis AX is referred as an “inner diameter side” and a side away from the bearing center axis AX is referred as an “outer diameter side”. Further, in the axial direction, a direction to which a diameter of a raceway surface **14** of an outer ring **12** decreases is referred as a “small diameter side”, and a direction to which the raceway surface **14** increases is referred as a “large diameter side”.

First Embodiment

[0068] As shown in FIG. **1**, a tapered roller bearing **1** of this embodiment includes an inner ring **11**, an outer ring **12**, a plurality of tapered rollers **15** interposed between the inner ring **11** and the outer ring **12**, and a retainer **17** retaining the plurality of tapered rollers **15** at fixed intervals. The retainer **17** has a pocket **16** opened in the radial direction, and the pocket **16** holds the tapered roller **15**.

[0069] The outer ring **12** has a raceway surface **14** on which the tapered roller **15** rolls on an inner circumference of the outer ring **12**. The inner ring **11** has a raceway surface **13** on which the tapered roller **15** rolls on an outer circumference of the inner ring **11**, and has a large flange portion **19** and a small flange portion **18** which come into contact with end surfaces of the tapered roller **15**

on opposite sides in the axial direction across the raceway surface **13**, respectively.

[0070] The tapered roller bearing **1** of this embodiment is a large tapered roller bearing, for example, used to support a main shaft of a wind power generator. In the tapered roller bearing **1** having such feature, an average diameter of the tapered roller **15** is 40 mm or larger and an outer diameter of the bearing **1** is 1 m or larger. However, the use of the tapered roller **1** is not limited to this.

[0071] FIG. **2** is a front view of an inner ring assembly **As** consisting of the inner ring **11**, the retainer **17** and the tapered roller **15** as viewed from the axial direction. As shown in FIG. **2**, the retainer **17** is a segment retainer including a plurality of retainer segments **20** divided in a circumferential direction. To be specific, the retainer **17** consists of the plurality of the retainer segments **20** connected by connectors **25** described below. The retainer segment **20** of this embodiment is made of resin. Specifically, the retainer segment **20** is made of Poly Ether Ketone that contains carbon fibers or glass fibers. However, a material of the retainer segment **20** is not limited to this.

[0072] FIG. **3** is a perspective view of the inner ring assembly **As**. FIG. **4** is a cross-sectional view of the retainer segment **20**. FIG. **5** is a top plan view of the retainer segment **20** viewed from an outer side in the radial direction. FIG. **6** is a perspective view of the retainer segment **20**. In fact, FIG. **3** shows only one retainer segment **20**. In the following description, the retainer segment **20** is sometimes referred to simply as a “segment **20**”.

[0073] The segment **20** includes, as shown in FIG. **5**, a large-diameter-side arc-shaped portion **22** and a small-diameter-side arc-shaped portion **21** extending circumferentially and a plurality of pillars **23** which connect the large-diameter-side arc-shaped portion **22** and the small-diameter-side arc-shaped portion **21**. Namely, the small-diameter-side arc-shaped portion **21** and the large-diameter-side arc-shaped portion **22** oppose to each other at a predetermined interval, and the plurality of pillars **23** are extended between the small-diameter-side arc-shaped portion **21** and the large-diameter-side arc-shaped portion **22**.

[0074] A space surrounded by two adjacent pillars **23**, the small-diameter-side arc-shaped portion **21**, and the large-diameter-side arc-shaped portion **22** constitutes the pocket **16** for containing the tapered roller **15**. In this embodiment, six pillars **23** are provided on one segment **20**, and five pockets **16** are arranged in the circumferential direction. However, the number of the pockets **16** is not limited to this, and it is required to arrange three or more pockets **16** in the circumferential direction on one segment **20**.

[0075] As shown in FIG. **4**, the pockets **16**, **16** located at opposite ends in the circumferential direction of the five pockets **16** have a first guide claw **24a** extending toward the inner diameter side from the pillar **23**, and the other three central pockets **16** have a second guide claw **24b** extending toward an outer diameter side **OS** from the pillar **23**. The first guide claw **24a** prevents the tapered roller **15** from slipping out to an inner diameter side **IS**, and the second guide claw **24b** prevents the tapered roller **15** from slipping out to the outer diameter side **OS**.

[0076] Roller guide surfaces **24aa**, **24ba** as inner peripheral surfaces of the first and second guide claws **24a**, **24b** have concave shapes curved along the shape of the tapered rollers **15**. With the guide claws **24a**, **24b**, the retainer **17** is guided by only the tapered roller **15** during operation. In other words, the tapered roller bearing **1** of this embodiment can achieve a roller guide.

[0077] The tapered rollers **15** can be inserted into the three central pockets **16** from the inner diameter side **IS** of the segment **20**. The second guide claw **24b** prevents the tapered rollers **15** from slipping out to the outer diameter side **OS**. Moreover, in the inner ring assembly **As** of FIG. **3**, since the raceway surface **13** of the inner ring is located on the inner diameter side, the tapered rollers **15** is also prevented from slipping out to the inner diameter side.

[0078] As shown in FIG. **4**, the tapered roller **15** can be inserted from the outer diameter side **OS** into the pocket **16** located at an end in the circumferential direction. In the inner ring assembly **As** of FIG. **3**, since the raceway surface **13** of the inner ring is located on the inner diameter side, the tapered rollers **15** is also prevented from slipping out to the inner diameter side. However, there is a

possibility that the tapered rollers **15** contained in the pockets **16** located at the opposite ends shown in FIG. **4** fall off to the outer diameter side OS. In this embodiment, as shown in FIG. **3**, falling-off preventers **32** are arranged along outer diameter surfaces **23a** of the pillars **23** located at opposite ends of the segment **20**. The falling-off preventer **32** prevents the tapered roller **15** from falling off to the outer diameter side.

[0079] The falling-off preventer **32** is made of, for example, resin or metal. However, a material of the falling-off preventer **32** is not limited to this. As shown in FIG. **6**, the falling-off preventer **32** has a pillar-shaped portion **32a** extending along an outer diameter surface of the segment **20**, a base portion **32b** extending along a large-diameter-side lateral-surface of the segment **20** and a segment engaging portion **32c** engaged with an inner diameter surface of the segment **20**.

[0080] In this embodiment, the pillar-shaped portion **32a** extends along an outer diameter surface **22a** of the large-diameter-side arc-shaped portion **22**, the outer diameter surface **23a** of the pillar **23** and an outer diameter surface **21a** of the small-diameter-side arc-shaped portion **21**. However, it is sufficient that the pillar-shaped portion **32a** extends at least along the outer diameter surface **22a** of the large-diameter-side arc-shaped portion **22**. The pillar-shaped portion **32a** prevents the tapered roller **15** (FIG. **3**) from falling off by abutting the pillar-shaped portion **32a** against the outer diameter surface **22a** of the large-diameter-side arc-shaped portion **22** and the tapered roller **15** (FIG. **3**). In this embodiment, a contact surface **32aa** of the pillar-shaped portion **32a** with the tapered roller **15** (FIG. **3**) has a concave shape curved along an outer shape of the tapered roller **15** (FIG. **3**).

[0081] The base portion **32b** abuts a large-diameter-side lateral-surface **22b** of the large-diameter-side arc-shaped portion **22**. The segment engaging portion **32c** is a claw-shaped member which extends from the base portion **32b** to the small diameter side and is engaged with an inner diameter surface **22c** of the large-diameter-side arc-shaped portion **22**.

[0082] As described above, the falling-off preventer **32** is engaged with the outer diameter surface **22a**, the large-diameter-side lateral-surface **22b** and the inner diameter surface **22c** of the large-diameter-side arc-shaped portion **22** of the segment **20**. Specifically, the pillar-shaped portion **32a** of the falling-off preventer **32** is engaged with the outer diameter surface **22a** of the large-diameter-side arc-shaped portion **22**, and the segment engaging portion **32c** is engaged with the inner diameter surface **22c** of the large-diameter-side arc-shaped portion **22**. Accordingly, movement of the falling-off preventer **32** in the radial direction is restricted.

[0083] The falling-off preventer **32** of this embodiment has a guide portion **35** with which the connector **25** is engaged. The guide portion **35** is provided on the base portion **32b** of the falling-off preventer **32**. To be specific, the guide portion **35** is provided on an opposite side to a surface in which the base portion **32b** of the falling-off preventer **32** abuts the segment **20**.

[0084] In this embodiment, the guide portion **35** is an engaging groove with which the connector **25** (FIG. **1**) is engaged. Specifically, the guide portion **35** is a groove opened outward in a radial direction. However, the shape of the guide portion **35** is not limited to this. For example, as shown in a variation of FIG. **7**, the guide portion **35** may be an insertion hole into which the connector **25** is inserted.

[0085] In addition, as shown in a variation of FIG. **8**, a width **W1** of an entrance of the groove-shaped guide portion **35** may be smaller than a width **W2** of the connector **25**. The width of the engaging groove (the guide portion) **35** other than the entrance is larger than the width **W2** of the connector **25** ($W1 < W2 < W3$). In this case, the guide portion **35** is elastically deformed so that the connector **25** is inserted into the engaging groove (the guide portion) **35**. Accordingly, the connector **25** is prevented from slipping out from the engaging groove (the guide portion) **35** when assembling.

[0086] In a variation of FIG. **9**, the falling-off preventer **32** includes an inner ring engaging portion **34** extending radially inward from the base portion **32b**. As shown in FIG. **10**, in a state where the falling-off preventer **32** is attached to the segment **20**, the inner ring engaging portion **34** is engaged

with a large end surface **11a** of the inner ring **11**. Accordingly, it is possible to prevent an inner diameter side of the segment **20** from spreading due to tightening of the connector **25**.

[0087] As shown in FIG. 5, a falling-off preventer guide portion **33** is provided on an end portion in the circumferential direction of the large-diameter-side arc-shaped portion **32** of the segment **20**. The falling-off preventer guide portion **33** suppresses the movement of the falling-off preventer **32** in the circumferential direction. In this embodiment, the falling-off preventer guide portion **33** is a protrusion protruding radially outward from the end portion in the circumferential direction of the outer diameter surface **22a** of the large-diameter-side arc-shaped portion **32** and a protrusion protruding radially inward from an end portion in the circumferential direction of the inner diameter surface **22c** of the large-diameter-side arc-shaped portion **32**. However, the shape of the falling-off preventer guide portion **33** is not limited to this.

[0088] As shown in FIG. 5 and FIG. 6, the falling-off preventer **32** can be inserted into and drawn from the segment **20** from the large end surface **11a** (FIG. 1) of the inner ring **11** along the falling-off preventer guide portion **33**. In other words, the falling-off preventer **32** can be attached to and detached from the segment **20**.

[0089] Large-diameter-side protruding portions **26** are provided on opposite end sides of the large-diameter-side lateral-surface **22b** of the large-diameter-side arc-shaped portion **22** of the segment **20**. The large-diameter-side protruding portion **26** protrudes in the axial direction from the large-diameter-side lateral-surface **22b** of the large-diameter-side arc-shaped portion **22**. The large-diameter-side protruding portion **26** includes an engaging portion **27** which is engaged with the connector **25**. In this embodiment, the engaging portion **27** is an engaging groove to which the connector **25** is fitted.

[0090] The large-diameter-side protruding portion **26** is arranged so as not to interfere with the falling-off preventer **32** and the falling-off preventer guide portion **33**. In this embodiment, in the inner ring assembly as shown in FIG. 3, the falling-off preventer **32** is attached between the large-diameter-side protruding portion **26** and the falling-off preventer guide portion **33** in the circumferential direction. In other words, the large-diameter-side protruding portion **26** is located in the circumferential direction side by side with the falling-off preventer **32**, and the engaging portion **27** (FIG. 5) of the large-diameter-side protruding portion **26** is continued to the guide portion **35** (FIG. 6) of the falling-off preventer **32**. Each retainer segment **20** and each falling-off preventer **32** are connected by engaging the connector **25** (FIG. 1) with the engaging portion **27** (FIG. 5) and the guide portion **35** (FIG. 6) so as to be tightened.

[0091] The connector **25** of FIG. 1 will be described. The connector **25** is removable with respect to the segment **20** and the falling-off preventer **32** (FIG. 3) and prevents the segment retainers **17** arranged in a ring shape from being broken up. The connector **25** is, for example, a wire. However, the connector **25** is not limited to a wire, but may be a belt or the other similar members.

[0092] In this embodiment, opposite end portions of the connector **25** is connected together by a fastening portion or a fastening member. By arranging the connector **25** in the vicinity of the large-diameter-side arc-shaped portion **22** of the retainer **17** shown in FIG. 5, a clearance is secured to make it possible to remove the connector **25** by a hand or a tool after assembling the bearing. In addition, it is preferable that the fastening portion or the fastening member is arranged between the large-diameter-side protruding portions **26** of the opposite ends of the large-diameter-side arc-shaped portion **22** of the segment **20**.

[0093] While the connector **25** may be a single body extending continuously, the connector **25** may be divided into a plurality of the connector segments **25**, and ends of each connector segment **25** may be connected together by the fastening portion and a fastening member **42** (FIG. 19). In this case, it is preferable that a plurality of the fastening portions and the fastening members **42** is arranged on a circumference at equal intervals. By the fastening portion or the fastening member **42** (FIG. 19), a fastening force can be equally applied to the entire connector **25**.

[0094] When a wire is used as the connector **25**, a hook or a turnbuckle (FIG. 19) can be used as

the fastening member **42**. Using a turnbuckle is preferable since a turnbuckle is removable, the tightening force will not decrease, and the tightening force is adjustable. When a wire is used as the connector **25**, it is preferable to use a removable buckle since the tightening force will not decrease. [0095] As shown in FIG. **19**, when a turnbuckle is used as the fastening member **42**, the fastening member **42** has a body having a female thread portion. Ends of the connectors **25** can be connected together by screwing a male thread portion provided on the end of the connector **25** into the body of the fastening member **42**. Rotating the body of the turnbuckle **42** can impart a tension to the connector **25**, and rotating the body of the turnbuckle **42** reversely makes it possible to unbind the opposite ends of the connectors **25**.

[0096] In this embodiment, as shown in FIG. **5**, the small-diameter-side arc-shaped portion **21** of the segment **20** includes a small-diameter-side protruding portion **28**. The small-diameter-side protruding portion **28** protrudes in the axial direction from a small-diameter-side lateral-surface of the small-diameter-side arc-shaped portion **21**. In this embodiment, the small-diameter-side protruding portions **28** are provided on opposite ends of the small-diameter-side arc-shaped portion **21** in the circumferential direction.

[0097] As shown in FIG. **11**, a ring-shaped jig **31** can be fitted into the small-diameter-side protruding portion **28**. When a large diameter side of each segment **20** is bound by the connectors **25**, the ring-shaped jig **31** fitted into the small-diameter-side protruding portion **28** prevents the inner diameter side of the segment **20** from spreading due to tightening of the connector **25**. The ring-shaped jig **31** is freely removable.

[0098] In a variation of FIG. **12**, instead of the small-diameter-side protruding portion **28** (FIG. **11**), a small-diameter-side-engaging portion **36** is provided on the small-diameter-side lateral-surface of the small-diameter-side arc-shaped shaped portion **21** of the retainer segment **20**. In the variation of FIG. **12**, the small-diameter-side engaging portion **36** is an engaging groove with which a small-diameter-side connector **37** is engaged. However, it is only required to engage the small-diameter-side connector **37** with the small-diameter-side engaging portion **36**. Therefore, the small-diameter-side engaging portion **36** is not limited to an engaging groove.

[0099] The small-diameter-side connector **37** is, for example, a wire and is removable with respect to the segment **20**. The retainer segments **20** may be connected by the small-diameter-side connector **37**. When the large diameter side of each segment **20** is bound by the connectors **25**, tightening of the small-diameter-side connector **37** fitted to the small-diameter-side engaging portion **36** prevents the inner diameter side of the segment **20** from spreading.

[0100] An assembling procedure of the tapered roller bearing of this embodiment will be described. Firstly, as shown in FIG. **3**, the inner ring assembly As in which the inner ring **11**, the tapered roller **15** and the retainer **17** are integrated is assembled. To assemble the inner ring assembly As, first of all, the tapered rollers **15** contained in the pockets **16** (FIG. **1**) other than opposite ends of the pockets **16** are arranged on the raceway surface of the inner ring **11** with the large end surface **11a** (FIG. **13**) of the inner ring **11** facing downward.

[0101] When the tapered rollers **15** are arranged on the raceway surface **13** of the inner ring **11** with the large end surface **11a** (FIG. **13**) of the inner ring **11** facing downward, there is a possibility that the tapered rollers **15** fall off from the inner ring **11** due to self-weight. To prevent the tapered rollers **15** from falling off from the inner ring **11**, as shown in FIG. **13**, the distance from a central axis AX1 of the inner ring **11** to the leading edge of the large flange portion **19** is set larger than the distance from the central axis AX1 of the inner ring **11** to a center of gravity C1 of the tapered roller **15**.

[0102] Specifically, in a state in which the tapered roller **15** is arranged on the raceway surface **13** of the inner ring **11** with the large end surface **11a** of the inner ring **11** facing downward, when an angle of the large flange portion **19** of the inner ring **11** with respect to a straight line that is perpendicular to the central axis AX1 of the inner ring **11** is defined as I, a chamfered width of the leading edge of the large flange portion **19** is defined as H, a distance from the central axis AX1 of

the inner ring **11** to the center of gravity **C1** of the tapered roller **15** is defined as $y1$, and a diameter of the large flange portion **19** is defined as J , the condition (1) below may be satisfied.

[00003] $(J/2) - H \cdot \cos l > y1$ (1)

[0103] Then, as shown in FIG. **14**, the tapered rollers **15** are inserted into opposite ends of the pockets **16** of the segment **20**. In this state, as shown in FIG. **15** and FIG. **16**, the segment **20** is put over the inner ring **11** in which the tapered rollers **15** are arranged. Consequently, the tapered rollers **15** are inserted from the inner diameter side into the pockets **16** other than opposite ends of the pockets **16**.

[0104] Subsequently, as shown in FIG. **17**, the falling-off preventer **32** is inserted from a large end surface side of the inner ring **11** along the falling-off preventer guide portions **33** arranged at opposite ends of the large-diameter-side arc-shaped portion **22** in order to prevent the tapered roller **15** from falling off toward the large diameter side.

[0105] Thereafter, the ring-shaped jig **31** (FIG. **19**) that is an L-shape is fitted into the small-diameter-side protruding portion **28** protruding toward a small diameter side of the segments **20** arranged in a ring shape shown in FIG. **18**. FIG. **18** shows a state before the ring-shaped jig **31** (FIG. **19**) is fitted, and FIG. **19** shows a state after the ring-shaped jig **31** is fitted.

[0106] Further, as shown FIG. **19**, the connector **25** is engaged with the engaging portion **27** of the large-diameter-side protruding portion **26** of the large-diameter-side lateral-surface **22b** (FIG. **5**) of the segment **20** and the guide portion **35** of the falling-off preventer **32**. Ends of the connectors **25** arranged at an outer diameter side of the segment **20** are bound by the fastening members **42**, and, in addition, the connectors **25** are tightened by the fastening members **42**. Accordingly, a plurality of the segments **20** arranged in the ring shape is integrated. As described above, the assembly of the inner ring assembly **As** is completed. The ring-shaped jig **31** can be removed by moving it along the central axis after the inner ring assembly **As** is assembled.

[0107] As shown in FIG. **22** and FIG. **23**, the inner ring assembly **As** is incorporated into the outer ring **12** after the inner ring assembly **As** is assembled. The inner ring assembly **As** is incorporated into the outer ring **12** in an upside-down state with a small diameter side of the inner ring assembly **As** facing downward. Even in the upside-down state, the segments **20** will not be broken up since outer circumferential sides of the segments **20** arranged in a ring shape are connected by the connectors **25**.

[0108] In addition, the tapered rollers **15** inserted into the pockets **16** other than opposite ends of the pockets **16** of each segment **20** are prevented from falling off by the second guide claw **24b** (FIG. **3**) provided on the outer diameter side, and the tapered rollers **15** inserted into opposite ends of the pockets **16** (FIG. **14**) are prevented from falling off by the falling-off preventer **32** (FIG. **17**). An incorporation into the inner ring **12** is performed after the ring-shaped jig **31** (FIG. **19**) is removed.

[0109] Even when the inner ring assembly **As** is turned into an upside-down state with the ring-shaped jig **31** (FIG. **19**) being removed, the tapered rollers **15** are fitted between the large flange portion **19** and the small flange portion **18**, so that the tapered rollers **15** are hooked on the small flange portion **18**. Consequently, the segments **20** are prevented from falling off.

[0110] When the tapered roller **15** is not in contact with the large flange portion **19**, the tapered roller **15** rotates in an arrow **AR1** about a point **A** on the raceway surface **13** as shown in FIG. **20**. When the tapered roller **15** is in contact with the large flange portion **19**, the tapered roller **15** rotates in an arrow **AR2** about a point **B** on an inner surface of the large flange portion **19** as shown in FIG. **21**.

[0111] When the tapered roller **15** shown in FIG. **21** rotates about the point **B**, an end surface of the tapered roller **15** on the small diameter side comes into contact with a point **C** on a side surface of the small flange portion **18**. Therefore, the rotation of the tapered roller **15** is suppressed. Here, the point **C** is a contact point at which the side surface of the small flange portion **18** comes into

contact with an end surface of the tapered roller **15** on the small diameter side when the tapered roller **15** is rotated around a leading edge of the large flange portion **18** of the inner ring **11**. In other words, the point C is an intersection of a spline curve SC drawn by a locus of a small end surface corner of the tapered roller **15** when the tapered roller **15** rotates, with the side surface of the small flange portion **18**.

[0112] Specifically, in a state in which the retainer segments **20** are connected by the connectors **25** shown in FIG. **22**, when a diameter of the small flange portion **18** of the inner ring **11** shown in FIG. **21** is defined as M, an angle of the small flange portion **18** with respect to a straight line that is perpendicular to the central axis AX1 of the inner ring **11** is defined as L, a chamfered width of a leading edge of the small flange portion **18** is defined as K, and a distance, to the central axis AX1, from the contact point C is defined as y3, the condition (2) below is satisfied.

[00004] $(M / 2) - K \cdot \text{Math. cos} L > y3$ (2)

[0113] After assembling the tapered roller bearing **1** shown in FIG. **23**, it is possible to remove the connector **25** and the falling-off preventer **32** by loosening the fastening member **42** (FIG. **19**). After assembling the tapered roller bearing **1**, each segment **20** will not be broken up even if the connectors **25** and the falling-off preventers **32** (FIG. **19**) are removed.

[0114] According to this embodiment, as described above, since the connector **25** and the falling-off preventer **32** shown in FIG. **19** are removable, the inner ring assembly As is integrated by the connector **25** and the falling-off preventer **32**, and the connector **25** and the falling-off preventer **32** can be removed after incorporating a bearing. Moreover, since the retainer **17** shown in FIG. **4** is guided by only the tapered roller **15** with the first guide claw **24a** and the second guide claw **24b** during operation, the roller guide can be realized.

Second Embodiment

[0115] A second embodiment of the present invention will be described. In the following description, the same features as in the first embodiment are designated by the same reference numerals, and the detailed description thereof may be omitted.

[0116] As shown in FIG. **24**, the tapered roller bearing **1** according to the second embodiment includes the inner ring **11**, the outer ring **12** concentrically arranged over the outer circumference of the inner ring **11**, the plurality of tapered rollers **15** arranged between the inner ring **11** and the outer ring **12**, and the retainer **17** including the pocket **16** that holds the tapered roller **15** at fixed intervals.

[0117] The outer ring **12** has the raceway surface **14** on which each tapered roller **15** rolls on the inner circumference of the outer ring **12**.

[0118] The inner ring **11** has the raceway surface **13** on which each tapered roller **15** rolls on the outer circumference of the inner ring **11**, and the large flange portion **19** and the small flange portion **18** which come into contact with end surfaces of the tapered roller **15** on the opposite sides in the axial direction across the raceway surface **13**.

[0119] In a large tapered roller bearing **1** used to support a main shaft or the like of a wind power generator, an average diameter of the tapered roller **15** is 40 mm or larger and an outer diameter of the bearing **1** m or larger.

[0120] The retainer **17** is a segment retainer consisting of the plurality of segments **20**, as shown in FIG. **25**.

[0121] As shown in FIG. **25**, FIG. **26** and FIG. **27**, the segment **20** has a pair of the small-diameter-side arc-shaped portion **21** and the large-diameter-side arc-shaped portion **22** opposing at a predetermined interval, and the plurality of pillars **23** extended between the small-diameter-side arc-shaped portion **21** and the large-diameter-side arc-shaped portion **22**. A space surrounded by two adjacent pillars **23**, the small-diameter-side arc-shaped portion **21**, and the large-diameter-side arc-shaped portion **22** is defined as the pocket **16** for containing the tapered roller **15**.

[0122] Three or more pockets **16** for containing the tapered rollers **15** are arranged in the

circumferential direction on one segment **20**. In this embodiment, six pillars **23** are provided on one segment **20**, and five pockets **16** for containing the tapered rollers **15** are arranged between the adjacent pillars **23**.

[0123] On an outer diameter side of the opposite pillars **23** of the pockets **16** of the five pockets **16** other than the pockets **16** located at ends in the circumferential direction i.e., three central pockets **16** in this embodiment, the guide claws **24** are provided for preventing the tapered rollers **15** from slipping out to the outer diameter side, so that the tapered rollers **15** can be inserted into the pockets **16** from the inner diameter side of the segment **20** as shown in FIG. 27.

[0124] The guide claws **24** are provided on an inner diameter side of the opposite pillars **23** of the pockets **16** located at ends in the circumferential direction, so that the tapered rollers **15** can be inserted into the pockets **16** from the outer diameter side. In addition, as shown in FIG. 26 and FIG. 27, the falling-off preventer **32** provided along the outer diameter surface of the pillar **23** located at opposite ends of the segment **20** prevents the tapered rollers **15** contained in opposite ends of the pockets **16** from falling off to the outer diameter side. The falling-off preventer **32** can be attached to and removed from the falling-off preventer fixing portion **33** formed at outer diameter surfaces of the opposed end sides, in the circumferential direction, of the small-diameter-side arc-shaped portion **21** and the large-diameter-side arc-shaped portion **22** from the large end surface side.

[0125] Since the guide claw **24** is provided on the inner diameter side of the pillar **23** of the pockets **16** located at ends in the circumferential direction of the pockets **16** retaining the tapered rollers **15**, the retainer **17** is of a rolling element guiding system by the tapered rollers **15** retained in the pockets **16** located at the ends.

[0126] As shown in FIG. 24 and FIG. 26, in a state in which the falling-off preventer **32** is inserted into the falling-off preventer fixing portion **33** provided on outer diameter surfaces of opposite ends of the small-diameter-side arc-shaped portion **21** and the large-diameter-side arc-shaped portion **22** from the large end surface side of the inner ring **11**, the inner ring engaging portion **34** engaged with a large end surface of the inner ring **11** is provided on an end, of the large end surface side of the inner ring **11**, of the falling-off preventer **32**.

[0127] A face being in contact with the tapered roller **15** of the falling-off preventer **32** is a round face.

[0128] Besides the falling-off preventer fixing portion **33** having a hole for inserting the falling-off preventer **32** as shown in FIG. 26 and FIG. 27, the falling-off preventer fixing portion **33** provided on the outer diameter surfaces of opposite ends of the small-diameter-side arc-shaped portion **21** and the large-diameter-side arc-shaped portion **22** may be a groove shape in which each falling-off preventer fixing portion **33** is opened to the adjacent segment **20** as shown FIG. 28. When the falling-off preventer fixing portion **33** is formed in the groove shape in which each falling-off preventer fixing portion **33** is opened as shown in FIG. 28, the falling-off preventer **32** (FIG. 26) as a common single body is inserted to the opposite falling-off preventer fixing portion **33** of the adjacent segments **20** formed in the groove shape, so that the tapered rollers **15** (FIG. 24) contained in opposite ends of the pockets **16** of the adjacent segments **20** can be prevented from falling off to the outer diameter side. Therefore, it is possible to reduce the number of components.

[0129] As shown in FIG. 26, the segments **20** are arranged in a ring shape with end surfaces of the small-diameter-side arc-shaped portion **21** and the large-diameter-side arc-shaped portion **22** being butted together.

[0130] On opposite end sides of the large-diameter-side lateral-surface of the large-diameter-side arc-shaped portion **22** of the segment **20**, the protruding portions **26** are provided so as to protrude in the axial direction, and each protruding portion **26** have the engaging portion **27** for the connector **25** (FIG. 24) to be arranged on the large-diameter side of the segments **20** arranged in a ring shape. Preferably, the protruding portions **26** located on opposite end sides of the large-diameter-side lateral-surface of the large-diameter-side arc-shaped portion **22** of the segment **20** are provided such that contact is avoided between the protruding portions **26** of the adjacent segments

20.

[0131] On the large-diameter-side lateral-surface of a structure **34**, engaged with a large end surface of the inner ring, of the falling-off preventer **32**, the groove-shaped guide portion **35** engaged with the connector **25** (FIG. **24**) wound around on the large diameter side of each segment **20** arranged in a ring shape is provided.

[0132] A wing portion **28** protruding to the small-diameter side on the small-diameter-side lateral-surface of the small-diameter-side arc-shaped portion **21** of the segment **20** is provided, and, as shown in FIG. **31**, the ring-shaped jig **31** is fitted into the small-diameter-side protruding portion of each segment **20** arranged in a ring shape when the large diameter side of each segment **20** arranged in a ring shape is bound by the connector **25** (FIG. **24**). Accordingly, the inner diameter side of the segment **20** is prevented from spreading due to tightening of the connector **25**.

[0133] The ring-shaped jig **31** is formed from a ring portion **31a** that abuts against the end surface on the small-diameter side of the inner ring **11** and an engaging portion **31b** provided so as to protrude toward the outer diameter side from an end of the ring portion **31a** in the axial direction, so that the engaging portion **31b** is fitted into the small-diameter-side protruding portion **28** of the segment **20** with the ring portion **31a** abutting against the end surface on the small-diameter side of the inner ring **11**.

[0134] The ends of the connector **25** wound around on the large diameter side of each segment **20** are bound by the fastening member **42** (FIG. **19**) such as a turnbuckle. When a turnbuckle is used as the fastening member **42** (FIG. **19**), the fastening member **42** has a body having a female thread portion. Ends of the connectors **25** can be connected together by screwing a male thread portion provided on the end of the connector **25** into the body of the turnbuckle **42**. Rotating the body of the turnbuckle can impart a tension to the connector **25**, and rotating the body in the opposite direction to the tightening direction makes it possible to unbind the opposite ends of the connectors **25**.

[0135] A wire or a belt can be used as the connector **25**. When a wire is used as the connector **25**, a removable hook or a turnbuckle can be used as the fastening member **42** (FIG. **19**). Using a turnbuckle is most preferable since a turnbuckle is removable, the tightening force will not decrease, and the tightening force is adjustable. When a belt is used as the connector **25**, it is preferable to use a removable buckle as the fastening member **42** (FIG. **19**) since the tightening force will not decrease.

[0136] The segment **20** is made of resin or steel. Poly Ether Ketone that contains carbon fibers or glass fibers can be used as resin that forms the segment **20**.

[0137] An assembling procedure of the tapered roller bearing **1** in which the segment **20** of the second embodiment is used is omitted since it is the same as that described in the first embodiment with FIGS. **13** to **23**.

[0138] The tapered roller bearing **1** of the second embodiment performs the same effect as the first embodiment.

Third Embodiment

[0139] In the second embodiment, the small-diameter-side protruding portion **28** protruding to the small diameter side on the small-diameter-side lateral-surface of the small-diameter-side arc-shaped portion **21** is provided, and, the ring-shaped jig **31** that is an L-shape is fitted into the small-diameter-side protruding portion **28** of each segment **20** arranged in a ring shape when the large diameter side of each segment **20** arranged in a ring shape is bound by the connector **25**. In a third embodiment shown FIG. **32**, instead of the ring-shaped jig **31**, the small-diameter-side engaging portion **36** is provided on the small-diameter-side lateral-surface of the small-diameter-side arc-shaped portion **21**, and the small-diameter-side engaging portion **36** is engaged with the small-diameter-side connector **37** wound around on the small diameter side of each segment **20** arranged in a ring shape. By removably binding ends of the small-diameter-side connector **37** with the fastening portion and a fastening member (not illustrated), the inner diameter side of the segment

20 is prevented from spreading due to tightening of the connector **25**.

Fourth Embodiment

[0140] In the second and third embodiments, the groove-shaped guide portion **35** engaged with the connector **25** is provided on the inner ring engaging portion **34**, engaged with the large end surface **11a** of the inner ring **11**, of the falling-off preventer **32**. On the other hand, in a fourth embodiment shown in FIG. **33**, the shape of the guide portion **35** can be a hole shape through which the connector **25** passes.

Fifth Embodiment: FIG. **34** to FIG. **38**

[0141] As shown in FIG. **34** to FIG. **36**, in each pocket **16**, each segment **20** may have a guide claw extending toward a direction opposite to a guide claw of the adjacent pocket **16**. Specifically, as shown in FIG. **36**, the pockets **16**, **16** located at the opposite ends in the circumferential direction and the pocket **16** located in the center in the circumferential direction have the second guide claws **24b** extending toward the outer diameter side OS from the pillars **23**. The two pockets **16**, **16** other than the pockets located at the opposite ends and in the center in the circumferential direction have the first guide claws **24a** extending toward the inner diameter side IS from the pillars **23**.

[0142] In the pockets **16**, **16** having the first guide claws **24a** extending toward the inner diameter side IS from the pillars **23**, the tapered rollers **15** are inserted from the outer diameter side OS of the segment **20**. In each pocket **16** having the second guide claws **24b** extending toward the outer diameter side OS from the pillars **23**, the tapered rollers **15** are inserted from the inner diameter side IS of the segment **20**. In each pocket **16** having the second guide claw **24b**, the second guide claw **24b** prevents the tapered rollers **15** from slipping out to the outer diameter side OS.

[0143] As shown in FIG. **35** and FIG. **37**, the falling-off preventer **32** prevents the tapered rollers **15** contained in the two pockets **16**, **16** having the first guide claws **24a** from falling off to the outer diameter side. As shown in FIG. **37** and FIG. **38**, the falling-off preventer **32** can be inserted into and drawn from the large end surface side of the inner ring along an outer diameter surface of the second guide claws in the central sides of the pockets **16**, **16** located at the opposite sides.

[0144] In addition to the structure of the first embodiment, the falling-off preventer **32** has a second segment engaging portion **32d** engaged with an end surface of the large-diameter-side protruding portion **26** of the segment **20**. As shown in FIG. **35** and FIG. **38**, the movement of the falling-off preventer **32** in the circumferential direction is suppressed by the tapered roller **15**, the second guide claws in the central sides of the pockets **16**, **16** located at the opposite sides, and the large-diameter-side protruding portion **26** engaged with the second segment engaging portion **32d**.

Accordingly, the falling-off preventer guide portion can be omitted, and thus it is possible to reduce the weight of the retainer segment **20** and improve lubricity of the tapered roller bearing compared to the other embodiments.

[0145] While preferred embodiments have been described thus far with reference to the drawings, various additions, modifications, or omissions can be made therein without departing from the principle of the present invention and are, thus, encompassed within the scope of the present invention.

REFERENCE SYMBOLS

[0146] **1** . . . tapered roller bearing [0147] **11** . . . inner ring [0148] **12** . . . outer ring [0149] **15** . . . tapered roller [0150] **16** . . . pocket [0151] **17** . . . retainer [0152] **20** . . . retainer segment (segment) [0153] **21** . . . small-diameter-side arc-shaped portion [0154] **22** . . . large-diameter-side arc-shaped portion [0155] **23** . . . pillar [0156] **24a** . . . first guide claw [0157] **24b** . . . second guide claw [0158] **25** . . . connector [0159] **26** . . . large-diameter-side protruding portion [0160] **27** . . . engaging portion [0161] **28** . . . small-diameter-side protruding portion [0162] **31** . . . ring-shaped jig [0163] **32** . . . falling-off preventer [0164] **33** . . . falling-off preventer guide portion [0165] **34** . . . inner ring engaging portion [0166] **35** . . . guide portion (insertion hole, engagement groove) [0167] **36** . . . small-diameter-side-engaging portion [0168] **37** . . . small-diameter-side connector

Claims

1. A tapered roller bearing comprising: an inner ring; an outer ring; a plurality of tapered rollers interposed between the inner ring and the outer ring; and a retainer which retains the tapered roller, the retainer including a plurality of retainer segments divided in a circumferential direction, the plurality of the retainer segments being connected by a removable connector, wherein each of the retainer segments has three or more pockets arranged in the circumferential direction which hold the tapered rollers, each of the retainer segments has a large-diameter-side arc-shaped part and a small-diameter-side arc-shaped portion extending circumferentially and a plurality of pillars which connect the small-diameter-side arc-shaped portion and the large-diameter-side arc-shaped portion, the pockets consisting of: the small-diameter-side arc-shaped portion; the large-diameter-side arc-shaped portion; and the pillar, a part of the pockets has a first guide claw extending toward an inner diameter side from the pillar, and the rest of the pockets has a second guide claw extending toward an outer diameter side from the pillar, further comprising a removable falling-off preventer which prevents the tapered roller contained in the pocket having the first guide claw from falling off toward the outer diameter side, wherein the falling-off preventer is engaged with an outer diameter surface and a large-diameter-side lateral-surface of the large-diameter-side arc-shaped portion of the retainer segment, and the retainer segments and the falling-off preventer are connected with the connector.
2. The tapered roller bearing as claimed in claim 1, wherein the falling-off preventer is arranged along the pillar or the second guide claw and is inserted from a large end surface side of the inner ring.
3. The tapered roller bearing as claimed in claim 1, wherein each of the retainer segments includes, on an end portion in the circumferential direction of the large-diameter-side arc-shaped portion, a falling-off preventer guide portion which suppresses the movement of the falling-off preventer in the circumferential direction.
4. The tapered roller bearing as claimed in claim 1, wherein the falling-off preventer has a guide portion with which the connector is engaged.
5. The tapered roller bearing as claimed in claim 4, wherein the guide portion is an insertion hole into which the connector is inserted or an engagement groove with which the connector is engaged.
6. The tapered roller bearing as claimed in claim 1, wherein the falling-off preventer has an inner ring engaging portion engaged with a large end surface of the inner ring.
7. The tapered roller bearing as claimed in claim 1, further comprising a large-diameter-side protruding portion which has an engaging portion with the connector and is disposed on opposite end sides of the large-diameter-side lateral-surface of the large-diameter-side arc-shaped portion of the retainer segment.
8. The tapered roller bearing as claimed in claim 1, wherein opposite ends of the connector are connected together by a fastening portion or a fastening member.
9. The tapered roller bearing as claimed in claim 1, wherein the connector is divided into a plurality of connector segments, and ends of each of the connector segments are connected together by a fastening portion or a fastening member.
10. The tapered roller bearing as claimed in claim 9, a plurality of the fastening portions or the fastening members is arranged at equal intervals in the circumferential direction.
11. The tapered roller bearing as claimed in claim 1, further comprising a small-diameter-side protruding portion, into which a ring-shaped jig can be fitted, which portion is arranged on a small-diameter-side lateral-surface of the small-diameter-side arc-shaped portion of the retainer segment.
12. The tapered roller bearing as claimed in claim 11, wherein the ring-shaped jig is freely removable.
13. The tapered roller bearing as claimed in claim 1, wherein the retainer segment is made of Poly

Ether Ketone that contains carbon fibers or glass fibers.

14. The tapered roller bearing as claimed in claim 1, wherein the falling-off preventer is made of resin or metal.

15. The tapered roller bearing as claimed in claim 1, wherein, in a state in which the tapered roller is arranged on a raceway surface of the inner ring with the large end surface of the inner ring facing downward, when an angle of a large flange portion of the inner ring with respect to a straight line that is perpendicular to a central axis of the inner ring is defined as I , a chamfered width of a leading edge of the large flange portion is defined as H , a distance from the central axis of the inner ring to a center of gravity of the tapered roller is defined as y_1 , and a diameter of the large flange portion is defined as J , the condition below is satisfied. $(J/2) - H \cdot \cos I > y_1$

16. The tapered roller bearing as claimed in claim 1, wherein, in a state in which the retainer segments are connected by the connector, when a diameter of the small flange portion of the inner ring is defined as M , an angle of the small flange portion with respect to a straight line that is perpendicular to a central axis of the inner ring is defined as L , a chamfered width of a leading edge of the small flange portion is defined as K , and a distance, to the central axis, from a contact point C at which a side surface of the small flange portion comes into contact with an end surface of the tapered roller on the small diameter side when the tapered roller is rotated around a leading edge of the large flange portion of the inner ring is defined as y_3 , the condition below is satisfied. $(M/2) - K \cdot \cos L > y_3$

17. The tapered roller bearing as claimed in claim 1, further comprising a small-diameter-side-engaging portion which removably engages a small-diameter-side-connector and is provided on the small-diameter-side lateral- surface of the small-diameter-side arc-shaped portion of the retainer segment.

18. The tapered roller bearing as claimed in claim 1, wherein the retainer is a retainer for a large tapered roller bearing whose outer diameter exceeds 1 m.
